

JARRELL I. REYES

FULL-STACK & MOBILE SOFTWARE ENGINEER

Quezon City, Metro Manila, Philippines • (+63) 09564330929 • jrllrey17@gmail.com • [imjajajah.github.io/my-portfolio](#)

Civil Service Professional Eligible

PROFESSIONAL SUMMARY

High-performing **Full-Stack and Mobile Software Engineer** with hands-on professional production experience and strong computer science fundamentals. Specialized in architecting multi-platform application suites, offline-first data synchronization protocols with SQLite/Flutter, transactional security with PayMongo webhooks, and complex relational SQL database query optimizations. Proven track record in delivering enterprise web administration systems and mobile field operation platforms.

CORE TECHNICAL COMPETENCIES

Languages & Web:	JavaScript (ES6+), TypeScript, PHP, Python, Dart, HTML5, CSS3, Tailwind CSS, Node.js, Express
Frameworks:	React, Flutter, Laravel, CodeIgniter MVC, Django
Databases & Storage:	PostgreSQL, MySQL, SQLite (Local DB Caching), Google Firebase (Firestore, Auth)
Integrations & Hardware:	RESTful APIs, Webhooks, PayMongo SDK, ESC/POS Bluetooth Thermal Printing, FCM Push Notifications
DevOps & Architecture:	Git, GitHub, Docker, Jira, Agile/Scrum, Role-Based Access Control (RBAC), Schema Design

PROFESSIONAL WORK EXPERIENCE

Freelance Full-Stack & Mobile Software Engineer	Aug 2025 – Present
Self-Employed • Quezon City, Philippines	
Ecosystem Architecture: Designed, deployed, and maintained multi-platform application suites for enterprise clients, encompassing admin web portals, field operational mobile apps, and customer-facing platforms serving thousands of active transaction logs. Offline Data Sync & Resilience: Engineered resilient, low-latency client-server synchronization protocols between local client SQLite databases and centralized backends, ensuring zero transaction loss during field offline usage. Security & Access Hardening: Implemented Webhook verification processors, multi-tier Role-Based Access Controls (RBAC), and atomic sequence ID generation logic to prevent unauthorized actions and auditing collisions.	

FEATURED ENTERPRISE & CLIENT PRODUCTION SYSTEMS

Tubig Pilipinas Enterprise Admin Web Portal Client: GT Industries	2025 – 2026
Stack: PHP (CodeIgniter MVC), MySQL, JavaScript, jQuery, PayMongo SDK, Excel (.xlsx) Parser	
Financial Ledger & Billing Engine: Re-engineered Statement of Account (SOA) rendering logic to enforce strict sequential compound billing calculations (Basic + VAT + Arrears + Penalties), completely eliminating legacy billing discrepancies. Thread-Safe Cashier Pipelines: Designed an isolated transaction-sequencing pipeline assigning distinct numeric ranges to cashier stations with reactive alerts before series exhaustion, guaranteeing zero auditing collisions. High-Volume Data Processing: Built a memory-efficient Excel (.xlsx) parsing engine featuring multi-column uniqueness constraints to support high-volume batch bill uploads without manual reconciliation.	
Tubig Pilipinas Field Meter Reader Mobile App Client: GT Industries	2025 – 2026
Stack: Flutter, Dart, SQLite (Local DB), RESTful APIs, ESC/POS Bluetooth Thermal Printing	
Offline-First Caching & Mutability: Architected client-side SQLite database structures enabling utility field teams to execute, cache, and update queued meter reads in real-time offline, enforced with photo-proof validations before server sync. Hardware Integration & Receipt Generation: Integrated ESC/POS Bluetooth thermal printer protocols to dynamically format and render standardized SOA bills on-site with 7-digit incremental reference tracking.	
Tubig Pilipinas Customer Self-Service App Client: GT Industries	2025 – 2026
Stack: Flutter, Dart, PayMongo SDK, RESTful APIs, iOS & Android	
Unified Ledger & Instant Settlements: Refactored mobile SOA layers to mirror standardized multi-tier fee structures and integrated PayMongo payment gateways for real-time bill settlements directly in-app.	

Mosang's Budbod Food Logistics & Ordering System

2025

Stack: Google Firebase (Firestore, Auth), PayMongo Webhook APIs, Third-Party Logistics APIs

- **Zero-Downtime Database Migration:** Reverse-engineered legacy database schemas and migrated live production assets to Google Firebase with zero downtime. Configured automated PayMongo webhook listeners for instant automated order updates.

TECHNICAL SYSTEMS & SOFTWARE PROJECTS

QR-Based Attendance & Tracking System

2025

Stack: JavaScript, Node.js, Express, HTML5 Camera Scan API, SQL / PostgreSQL

- **Real-Time Camera Scanning & Logging:** Developed an automated attendance tracking platform utilizing browser camera frame scanning, instant QR payload parsing, and high-concurrency database logging.

EDUCATION & ACADEMIC LEADERSHIP

Polytechnic University of the Philippines (PUP)

Graduated 2025

Bachelor of Science in Information Technology (BSIT) • Quezon City, Philippines

- **Academic Intern (Project Lead & Full-Stack Developer):** Spearheaded system architecture, schema modeling, and full-stack development for PUP academic administrative portals (Feb 2025 – Jun 2025).
- **Capstone Project Manager & Lead UI Developer:** Directed requirements modeling, task delegation, and Agile backlogs for cross-platform mobile/web software applications (Aug 2024 – Feb 2025).

CERTIFICATIONS

- **Responsive Web Design & JavaScript Algorithms and Data Structures** – freeCodeCamp (2025)
- **Google Project Initiation & Planning Certification** – Coursera (2024)
- **Fortinet Cybersecurity Foundations Certification** – Fortinet Training Institute (2024)